

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

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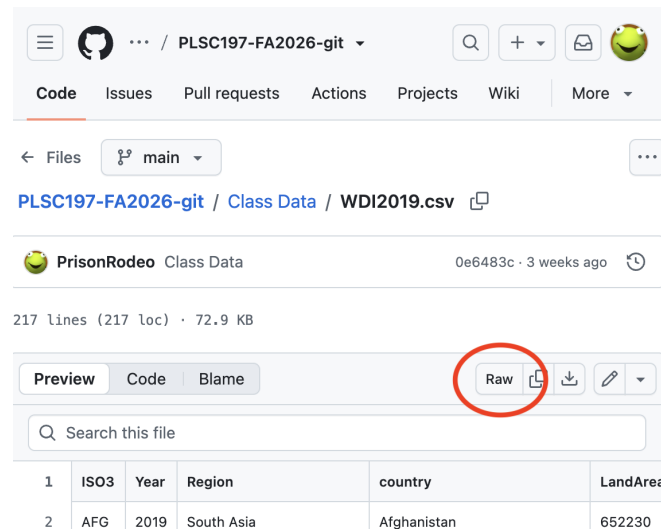
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World Development Indicators Data

Introduction

For our first applied lab session, we'll be visualizing data from the *World Development Indicators* (WDI) to illustrate key course models and concepts. The WDI are data collected on around 1,400 demographic, political, social, and economic indicators collected annually by the World Bank. Data are collected at the national level, for each of around 215 countries in the international system. The length of time each variable is collected varies significantly; the longest series extend back to 1960, while others are only available for very recent years or at discrete time points. (The World Bank also gathers and publishes WDI data on regional groupings, but for our purposes we'll focus on data at the national level.)

Detailed information about the WDI data (as well as bulk download and some on-line analysis) is available at the [WDI website](#). We'll be focusing on a subset of the whole WDI data, one containing data from only one year (2019), and roughly 40 variables.¹ Those data are in a file called `WDI2019.csv`, which is available in the course Github repo, in the folder labeled "Class Data." To download those data, navigate to the `.CSV` file on the repo, and click on the "Raw" button in the upper right of the spreadsheet view:



Depending on your browser, you'll either see the data in raw form (which you can download) or it will download the `.csv` file directly to your computer. Once you've downloaded the data, you can upload it into

¹While you need not concern yourself with this fact, the R code for obtaining and creating the WDI data for this lab (and for creating the figures in Monday's slides) is available on the course [Github repository](#), and makes use of the very useful `WDI` and `countrycode` packages, created by [Vincent Arel-Bundock](#) and his collaborators.

one of the no-code visualization tools (as we'll demonstrate in class), or to Google Sheets, or to an LLM to create visualizations.

The following are some brief descriptions of the variables available in the data, in roughly the order in which they appear in the `.csv` file. Longer / more detailed descriptions can be found with a basic web search.

Identifier Variables

- `ISO3` - The country's International Standards Organization (ISO) three-letter identification code (e.g., `CHE` for Switzerland).
- `Year` - The year that row of data applies to. The combination of `ISO3` and `Year` uniquely identifies every observation in the data.
- `Region` - The geographical region of the country. There are seven regions specified: East Asia & the Pacific, Europe & Central Asia, Latin America & the Caribbean, the Middle East & North Africa, North America, South Asia, and Sub-Saharan Africa.
- `country` - The name of the country (useful for labeling, etc.).

Continuous Variables

- `LandArea` - Land area (sq. km).
- `ArablePercent` - Arable Land (percent of total land area).
- `Population` - Population.
- `PopGrowth` - Population Growth (percent).
- `RuralPopulation` - Rural Population (percent of total).
- `UrbanPopulation` - Urban Population (percent of total).
- `BirthRatePer1K` - Birth Rate (births per 1K people).
- `FertilityRate` - Fertility Rate (births per woman).
- `PrimarySchoolAge` - Primary school starting age (years).
- `LifeExpectancy` - Life Expectancy at birth (years).
- `AgeDepRatioOld` - [Age Dependency Ratio](#) (old), percent of the working age population.
- `CO2Emissions` - CO2 Emissions (metric tons per capita).
- `GDP` - GDP (constant 2015 \$US).
- `GDPPerCapita` - GDP per capita (constant 2015 \$US).
- `GDPPerCapGrowth` - GDP Per Capita Growth (percent annual).
- `Inflation` - Inflation (CPI, annual percent).

- TotalTrade - Total Trade (percent of GDP).
- Exports - Exports (percent of GDP).
- Imports - Imports (percent of GDP).
- FDIIn - Inward Foreign Direct Investment (FDI) (percent of GDP).
- AgriEmployment - Percent of total employment in agriculture.
- MobileCellSubscriptions - Mobile / cellular subscriptions per 100 people.
- NaturalResourceRents - Total natural resource rents (percent of GDP).
- GovtExpenditures - Government Expenditures (percent of GDP).
- PublicHealthExpend - Public expenditure on health (percent of GDP).
- PoliticalStability - **Political Stability Score** (range $\approx$ [21,96], higher values = greater stability).
- RuleOfLaw - **Rule of Law Score** (range $\approx$ [17,93]; higher values = stronger rule of law).
- ControlOfCorruption - **Control of Corruption Score** (range $\approx$ [10,94]; higher values = greater control of corruption).

Discrete Variables

- IncomeLevel - WDI's four-category ordinal income level variable (*Low Income, Lower Middle Income, Upper Middle Income, High Income*)
- RuralLevel - Coded *High* if RuralPopulation is above the overall median, and *Low* if it is not.
- UrbanLevel - Coded *High* if UrbanPopulation is above the overall median, and *Low* if it is not.
- PopGrowthLevel - Coded *High* if PopGrowth is above the overall median, and *Low* if it is not.
- EconGrowthLevel - Coded *High* if GDPPerCapGrowth is above the overall median, and *Low* if it is not.
- TradeLevel - Coded *High* if TotalTrade is above the overall median, and *Low* if it is not.
- AgricultureLevel - Coded *High* if AgriEmployment is above the overall median, and *Low* if it is not.